

Curated Research Articles

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- **Defect Topology Governs Lithiation Pathways in Graphite Anodes** — score: 1.000 The study reveals that the defect topology of graphite, including specific types of stacking faults and rhombohedral intergrowths, significantly influences lithium insertion pathways in graphite anodes during battery operation. Lithium preferentially intercalates in defect-rich environments rather than in ideal 2H structures, highlighting the importance of defect characterization for understanding electrochemical performance and improving battery efficiency. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Tracking Dynamic Sulfur Electrochemistry by Operando Techniques in Alkali Metal-Sulfur Batteries** — score: 1.000 The article reviews advancements in operando techniques for studying the sulfur electrochemistry in alkali metal-sulfur batteries (AMSBs), such as Li-S, Na-S, and K-S. It emphasizes the importance of real-time analyses in understanding reaction pathways and interfacial dynamics, addressing challenges like polysulfide behavior and sluggish kinetics, while proposing strategies for future battery design through the integration of experimental methods and theoretical modeling. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Nanoscale Mapping of Transition Metal Ordering in Individual LiNi_{0.5}Mn_{1.5}O₄ Particles Using 4D-STEM** — score: 1.000 The study employs 4D-Scanning Transmission Electron Microscopy (4D-STEM) to analyze the transition metal ordering in individual LiNi_{0.5}Mn_{1.5}O₄ particles, revealing distinct ordered and disordered structures that significantly impact electrochemical performance in lithium-ion batteries. By developing a quantification method for local ordering, the researchers demonstrate that the degree of transition metal ordering varies throughout the particles and is influenced by annealing conditions, thus providing insights into optimizing battery materials. Journal: *Wiley: Small Methods: Table of Contents*
- **The First Electrochemical Cycle: State-of-Charge Dependent Formation and Evolution of the Solid Electrolyte Interphase on Hard Carbon Anodes in Sodium-Ion Batteries** — score: 1.000 This study investigates the formation and evolution of the solid electrolyte interphase (SEI) on hard carbon anodes during the first electrochemical cycle in sodium-ion batteries. Through a combination of advanced microscopy and spectroscopy techniques, the research reveals how the morphology and chemistry of the SEI change in response to varying states of charge, highlighting the transition from an oxide-rich layer to a NaF-dominated interphase and demonstrating significant alterations in surface conductivity during sodiation and desodiation. Journal: *Wiley: Small Methods: Table of Contents*
- **Chemomechanically Adaptive MXene–Si–Ag Dual-Seed Interlayer Enabling Low-Pressure Sulfide all-Solid-State Batteries** — score: 1.000 The article presents a novel MXene–Si–Ag hybrid interlayer designed to enhance the stability of sulfide-based all-solid-state batteries (ASSBs) under low stack pressure. By integrating immobile and mobile lithiophilic sites within a flexible MXene framework, the dual-seed structure improves lithium-ion transport and maintains robust solid-solid contact, enabling long cycling life even at ultralow pressures. This advancement supports the development of high-performance, pressure-tolerant ASSBs, achieving superior capacity retention across numerous cycles. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Electron Spin-State Regulation Unlocks Structural Robustness and Stable Sodium Storage in Na₄Fe₃(PO₄)₂P₂O₇ Cathode** — score: 1.000 The article discusses the development of a spin-state-engineered sodium-ion battery cathode, Na₄Fe_{2.94}Mo_{0.03}Co_{0.03}(PO₄)₂P₂O₇ (MC-NFPP), which improves electronic conductivity and structural stability through a transition of iron from an intermediate-spin to a high-spin state. This modification enhances charge transport, reduces Na⁺ migration barriers, and ensures robust cycling performance, achieving a remarkable capacity retention after extensive cycling, thereby addressing the limitations of traditional polyanionic cathodes for practical battery applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Reveal Non-Uniform Passivation on Lithium Metal Anode With Operando Spectroscopic**

- Ellipsometry** — score: 1.000 This study employs operando spectroscopic ellipsometry to investigate the non-uniform growth of passivation layers on lithium metal anodes in lithium metal batteries. By analyzing nucleation behaviors and crystallographic orientations of lithium (specifically Li(110) and Li(200)), the research reveals how these factors contribute to varying thicknesses of passivation films, offering insights for enhancing lithium interface stability and manufacturing consistency. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Tailoring Electrolyte Chemistry Through Precise Organic Synthesis for Lithium Metal Batteries** — score: 0.900 The article discusses how precise organic synthesis can enhance the electrolyte chemistry of lithium metal batteries, which are crucial for achieving high energy densities exceeding 500 Wh kg⁻¹. By carefully manipulating molecular properties, researchers can improve solvation behavior and interphase stability, thus overcoming conventional electrolyte limitations. The review emphasizes a systematic approach to battery optimization that addresses both performance and sustainability concerns, aiming to facilitate the transition of these innovations into practical applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
 - **Multi-Cation Synergy in Medium-Entropy Spinel Oxide for Practical Zn-Ion Pouch Cells** — score: 0.800 The study presents a novel multi-cation medium-entropy spinel oxide cathode, Zn_{0.86}Co_{0.28}Ni_{0.04}Cu_{0.03}Mn₂O₄ (ZNCNMO), which addresses challenges in manganese oxide cathodes for aqueous zinc-ion batteries by preventing dissolution and structural issues. This cathode achieves an impressive capacity of 520 mAh g⁻¹ and maintains 98.3% of its capacity after 2600 cycles, showcasing its potential for sustainable energy applications with a practical energy density of 56.3 Wh kg⁻¹. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
 - **Enabling high-voltage and wide-temperature quasi-solid-state batteries via spatially engineered polyether electrolyte interfaces** — score: 0.800 The article discusses advancements in quasi-solid-state batteries that utilize spatially engineered polyether electrolyte interfaces to enhance performance across high-voltage and wide-temperature ranges. The research demonstrates that optimizing these interfaces can significantly improve battery efficiency and stability, potentially leading to more reliable energy storage solutions. Journal: *Joule*
 - **Topological Defect-Engineered Cu/N-C Catalysts for Zinc-Air Battery With Ultra-Stable Cycling Over 2000 h at High Current Densities** — score: 0.800 The article presents a novel topological defect-engineered copper single-atom catalyst integrated with nitrogen-doped carbon nanofibers for use in zinc-air batteries. This innovative approach stabilizes the active Cu-N_x sites, enhancing their durability and optimizing oxygen adsorption, resulting in an exceptional cycling stability of over 2000 hours at high current densities, far surpassing current benchmarks. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
 - **Multifunctional Diluent Enables High-Voltage and Flame-Retardant Potassium-Ion Batteries** — score: 0.800 This study presents a novel non-flammable multifunctional diluent, ethoxy(pentafluoro)cyclotriphosphazene (PFPN), which enhances the performance and safety of potassium-ion batteries. By optimizing the electrode-electrolyte interphase and improving ionic transport, the PFPN-based electrolyte enables stable long-term cycling performance, supporting high-voltage operations up to 4.5 V across various cell configurations. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
 - **Synergistic Size and Electronic Engineering of Cr₃C₂@C(N_x) Nanoparticles Via Arc-Discharge for High-Performance Zn-Air Batteries** — score: 0.800 The article details the innovative synthesis of core-shell Cr₃C₂@C nanoparticles via DC arc-discharge plasma, which allows for precise control over size and structure through cooling techniques. The resultant nitrogen-doped catalysts demonstrate superior oxygen reduction reaction (ORR) performance in zinc-air batteries, outperforming conventional commercial Pt/C catalysts, and provide insights into their mechanisms via in situ optical emission spectroscopy. This study highlights a new approach to enhance the efficacy of non-precious electrocatalysts by strategically manipulating size and electronic properties. Journal: *Wiley: Small Methods: Table of Contents*

- **TiO₂-Assisted Derivation of Silicon Quantum Dots From Silica Cages Achieving 0.04% Capacity Decay Per Cycle to 900 Cycles** — score: 0.800 The article presents a TiO₂-assisted method for deriving silicon quantum dots from silica cages, which significantly improves the cycling stability of the anode by minimizing capacity decay to just 0.04% per cycle over 900 cycles. This approach allows for nearly 100% initial Coulombic efficiency and supports a full cell with LiFePO₄ that achieves an impressive energy density of 362.4 Wh · kg⁻¹ after 400 cycles, demonstrating enhanced performance and longevity in energy storage applications. Journal: *Wiley: Small Methods: Table of Contents*
- **Mechanically Engineered Wood Hard Carbon Anodes Achieving 94% Initial Coulombic Efficiency for High Performance Sodium-Ion Batteries** — score: 0.800 The study presents a novel mechanical pretreatment method for basswood, resulting in hard carbon anodes with an optimized microstructure that achieves a high initial Coulombic efficiency of 94.1% and a reversible capacity of 324 mAh g⁻¹ at 0.1 C. This advancement enhances sodium ion storage performance through improved low-voltage Na⁺ storage and reduced electrolyte decomposition, with simulations further supporting the favorable electrochemical properties of the modified material. Journal: *Wiley: Small Methods: Table of Contents*
- **Ultrahigh-Loading Cathodes for Next-Generation Lithium-Ion Batteries** — score: 0.800 The article discusses the potential of ultrahigh-loading cathodes to enhance the energy density of lithium-ion batteries, while also highlighting the challenges posed by increased electrode thickness such as ion transport issues, mechanical stress, and degradation mechanisms. It reviews various cathode materials and their unique degradation behaviors, evaluates strategies for improving electrode performance, and suggests future research directions to overcome these obstacles and improve industrial feasibility. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Redox-Catalytic p-Toluquinone With Dynamic Proton-Iodine Modulation for High-Capacity and Durable Zn-I₂ Battery** — score: 0.800 The article presents a novel redox-catalytic cathode made from bio-derived p-toluquinone confined to porous activated carbon (TQ@AC), which enhances the performance of Zn-I₂ batteries by improving iodine conversion and reducing polyiodide shuttling. This cathode exhibits a high capacity of 410 mAh g⁻¹ and maintains 85% capacity retention over 63,000 cycles, demonstrating excellent stability and operational efficiency across a wide temperature range, thus showcasing its potential for sustainable energy storage solutions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Surface-Fluorinated Conductive Carbon for Coupled Electron/Ion Transport Through the Graphite Anode** — score: 0.800 The article presents a novel gas-phase fluorination method that transforms conventional Super P conductive carbon into a surface-fluorinated variant (F-SP), enhancing both electron and lithium ion transport in graphite anodes for lithium-ion batteries. This modification results in improved charge transfer kinetics, reduced interfacial polarization, and robust cycling stability, allowing full cells to maintain 81.99% capacity after 500 cycles, thereby significantly enhancing fast-charging capabilities. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Hard Carbon Networks for Mitigating Graphite Lattice Strain Under Extreme-Low-Temperature Fast Charging** — score: 0.800 The article presents a novel strategy involving a hard carbon percolating network integrated within graphite anodes to enhance the performance of lithium-ion batteries during extreme low-temperature fast charging. This approach effectively redistributes ionic flow and minimizes mechanical strain, resulting in impressive cycle stability and capacity retention even at very low temperatures, positioning it as a viable solution for high-power electric vehicle batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Comprehensive Spectroscopic Elucidation of Atomically Dispersed FeN₄ Active Sites of Fe-N-C Electrocatalysts** — score: 0.700 This article investigates the atomically dispersed active sites of Fe-N-C electrocatalysts using advanced spectroscopic techniques, revealing that the prominent active sites are FeN₄C₁₀OH and FeN₄C₁₂OH. By employing valence-to-core X-ray emission spectroscopy alongside theoretical modeling, the research enhances the understanding of the structural characteristics and effectiveness of these catalysts for oxygen reduction reactions, paving the way for insights into other

single-atom catalysts as well. Journal: *Wiley: Small Methods: Table of Contents*

- **Phosphorus-mediated engineering of topological carbon rings for accelerated bulk diffusion in hard carbon toward high-performance low-temperature sodium-ion batteries** — score: 0.600 The article explores how incorporating phosphorus into the design of topological carbon rings can enhance the bulk diffusion of sodium ions in hard carbon materials. This advancement aims to improve the performance of low-temperature sodium-ion batteries, highlighting the potential for better energy storage solutions. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Small congeneric solvents for practical sodium metal batteries** — score: 0.600 The study examines how the size of congeneric solvents influences the performance of sodium metal batteries, revealing that smaller solvents enhance fast-cycling capabilities, while demonstrating similar behavior at lower rates. Using generative screening methods, the research identifies DMFSA as a promising electrolyte that achieves high Coulombic efficiency and oxidative stability. This approach may be applicable to a wider range of electrolytes and metal-electrode systems. Journal: *Joule*
- **Dry-coated ZIF-8 interlayers regulate solvation and interphase chemistry for safer high-voltage Ni-rich cathodes** — score: 0.600 The article discusses the development of dry-coated ZIF-8 interlayers that enhance solvation processes and interphase chemistry, thereby improving the safety and performance of high-voltage Ni-rich cathodes in energy storage applications. The research highlights the potential benefits of these interlayers in mitigating challenges associated with high-voltage battery systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Iron-Driven Internal Percolation Enables Topological Construction of 3D Interconnected Porous Si@C Anodes** — score: 0.600 This research introduces an innovative method for creating porous Si@C anodes using an iron-driven internal percolation approach, which enhances the stability of the structure while accelerating the reduction of SiO₂. The resulting anode demonstrates impressive performance characteristics, achieving a high reversible capacity and maintaining excellent cycling stability, thus providing a promising technique for the design of advanced porous electrodes in lithium-ion batteries. Journal: *Wiley: Small Methods: Table of Contents*
- **High-Entropy Enhanced d-p Orbital Hybridization in Na₄Fe₃(PO₄)₂P₂O₇ Cathode: Electronic Structure Engineering** — score: 0.400 The article discusses the enhancement of electronic properties in the cathode material Na₄Fe₃(PO₄)₂P₂O₇ through high-entropy engineering, specifically focusing on the role of enhanced d-p orbital hybridization. This innovative approach aims to improve the material's performance in energy storage applications. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Continuous, fuel-cell-like operation of aluminum-air batteries via precipitate control and interfacial hydration** — score: 0.300 The study by Shi et al. demonstrates that by managing discharge-product accumulation and enhancing interfacial hydration, aluminum-air batteries can achieve continuous operation akin to fuel cells. Their innovative design incorporates a replaceable anode pouch and electrolyte microdosing, allowing for sustained energy output and reuse of components, thus laying the groundwork for efficient metal-air powered systems. Journal: *Joule*
- **The importance of fit-for-purpose structure characterization in autonomous laboratories** — score: 0.300 The article discusses the critical role of fit-for-purpose structure characterization in the context of autonomous laboratories, where advances in robotic experimentation are transforming material synthesis. It emphasizes that the effectiveness of autonomous systems hinges on structured measurement techniques that provide essential insights at the atomic level, advocating for adaptive approaches to enhance materials discovery. Journal: *Nature Reviews Materials*
- **Proton-transfer flash etching in dilute O₂-H₂O vapor: ‘molecular scissors’ for converting waste pyrolytic carbon black into sustainable hard carbon anodes** — score: 0.200 The article discusses a novel method involving proton-transfer flash etching in a dilute O₂-H₂O vapor, which acts as ‘molecular scissors’ to transform waste pyrolytic carbon black into sustainable hard carbon anodes. This innovative approach aims to enhance the efficiency of energy storage materials. Journal: *ScienceDirect Publication: Energy Storage Materials*

- **Origin of elastic anisotropy in O3-type layered sodium cathode** — score: 0.200 The article explores the mechanisms behind elastic anisotropy in O3-type layered sodium cathodes, focusing on the structural and compositional factors that contribute to this unique property. The findings advance the understanding of material behavior in energy storage applications, providing insights that could enhance the performance of sodium-ion batteries. Journal: *ScienceDirect Publication: Nano Energy*
- **Local Strain Switches CO Reduction to a 3.13 V Li C O Pathway for Li-CO₂ Batteries** — score: 0.200 The article explores how local strain in Li-CO₂ batteries can alter the electrochemical pathway for CO₂ reduction, specifically directing it towards a more efficient 3.13 V Li C O route. This finding could have significant implications for enhancing the performance of energy storage systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Deciphering Thermal Failure in Lithium Metal Batteries: Mechanism Elucidation and Mitigation Strategies** — score: 0.200 The article investigates the causes of thermal failure in lithium metal batteries, providing insights into the underlying mechanisms and proposing strategies for mitigation. By analyzing these failures, the authors aim to enhance the safety and performance of battery systems, contributing to the advancement of energy storage technologies. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Microwave-assisted solvo-metallurgical recovery of lithium and transition metals from lithium-ion battery waste** — score: 0.200 The article discusses a novel microwave-assisted solvo-metallurgical technique for efficiently recovering lithium and transition metals from lithium-ion battery waste. This method aims to enhance resource recovery while minimizing environmental impacts, presenting a promising approach for sustainable recycling in the battery industry. Journal: *ScienceDirect Publication: Nano Energy*
- **Boosting PEMWE Performance Via the Local Electronic Regulation of Ir_{0.5}Ru_{0.5}/NbN Catalyst With Synergistic Vacancy and a Doping Engineering Strategy** — score: 0.200 This article presents a novel approach to enhance the performance of a proton exchange membrane water electrolysis (PEMWE) catalyst by utilizing defect engineering on a supported Ir_{0.5}Ru_{0.5} nanocluster. The study demonstrates that by modulating the electron cloud density and strengthening metal-support interactions through doping, the catalyst achieves high efficiency and stability for oxygen evolution reactions, showcasing a significant reduction in iridium usage while maintaining robust hydrogen production capabilities. Journal: *Wiley: Small Methods: Table of Contents*
- **Manipulating Valence-Heterogeneous Pt Dual Sites via Metal-Support Interactions for Highly Efficient Alkaline Hydrogen Evolution** — score: 0.200 The study presents a method for enhancing the efficiency of alkaline hydrogen evolution by fabricating low-loading Pt nanoclusters on carbon nanotubes, which promotes strong metal-support interactions. By manipulating the ratio of valence-heterogeneous Pt sites through nitrogen doping and the carbon support's work function, the researchers achieved a remarkably low overpotential and high durability for the catalyst, providing valuable insights for future sustainable hydrogen production. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Solar-Driven Upcycling of Waste Polyethylene Terephthalate** — score: 0.000 This review examines advancements in solar-driven upcycling of waste polyethylene terephthalate (PET), focusing on methods such as photocatalysis and photothermal processes. It details catalyst engineering and mechanisms for improving efficiency and product selectivity while addressing the integration of these technologies into scalable solutions for enhancing the recycling of plastics and promoting a circular economy. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Al-Induced Cu²⁺-Cu⁺ Polarization in Ampere-Level Waste-Treating-Waste Electrocatalysis for Concurrent CO₂-to-C₂ Production and Plastic Upcycling** — score: 0.000 This study presents a novel electrocatalytic strategy that utilizes an AlCuIn nanorod/N-doped graphene oxide catalyst to simultaneously upcycle polyethylene terephthalate (PET) and reduce CO₂ to value-added chemicals. The research demonstrates that the Al-induced polarization effect enhances catalytic activation and product selectivity, achieving high faradaic efficiencies and stability in long-duration oper-

ations, thus offering a promising solution to plastic waste and carbon emissions through sustainable electrochemical processes. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Author Correction: Durable all-inorganic perovskite tandem photovoltaics** — score: 0.000 The article presents a correction to a previous publication regarding all-inorganic perovskite tandem photovoltaics, focusing on their durability. It highlights updates or clarifications made to the original findings and methodologies, contributing to the advancement of this promising solar technology. Journal: *Nature*
- **Author Correction: Cooperation conflicts with equality when allocating public goods** — score: 0.000 The article provides an author correction addressing findings related to the interplay between cooperation and equality in the context of public goods allocation. It highlights how efforts to foster cooperative behavior among individuals can sometimes lead to unequal distributions, thereby challenging the principles of fairness in resource sharing. Journal: *Nature*
- **Learning millisecond protein dynamics from what is missing in NMR spectra** — score: 0.000 The article discusses a novel approach to understanding millisecond-scale protein dynamics by analyzing gaps in NMR spectral data. This method leverages advanced computational techniques to infer dynamic behaviors that are typically obscured, providing deeper insights into protein function and structure. Journal: *Nature*
- **Author Correction: Nucleolar URB1 ensures 3 ETS rRNA removal to prevent exosome surveillance** — score: 0.000 This article presents a correction to prior findings regarding the protein URB1, which is crucial for the effective removal of the 3' external transcribed spacer (ETS) from ribosomal RNA (rRNA). By ensuring proper rRNA processing, URB1 plays a vital role in preventing unwanted exosomal degradation of rRNA transcripts. Journal: *Nature*
- **How and when to use artificial intelligence in your science job application** — score: 0.000 The article discusses insights from a recent webinar aimed at both hiring managers and job seekers on effectively incorporating artificial intelligence into the science job application process. It provides guidance on the optimal use of AI tools to enhance applications and interview strategies within the scientific field. Journal: *Nature*
- **Where are you ticklish? Researchers map the sensation across cultures** — score: 0.000 Research has identified consistent “tickle hotspots” across Chinese, Dutch, and Greek cultures, revealing that individuals are generally more ticklish in areas of the body that are infrequently touched. This study highlights the universal nature of ticklishness despite cultural differences. Journal: *Nature*
- **I caught my students using AI to cheat in an exam — here’s what universities must do to stamp this out** — score: 0.000 The article emphasizes the growing challenge of AI-assisted cheating in higher education, arguing that without robust policies and effective monitoring, the integrity of academic standards could be compromised. It calls for universities to implement proactive measures to combat this rising issue. Journal: *Nature*
- **What to do when you have too many PhD projects on the go** — score: 0.000 Zachary Sentell shares his strategy for successfully managing multiple PhD projects by prioritizing and narrowing his focus, which enabled him to publish several papers and graduate on schedule. The article offers insights into effective time management and decision-making for PhD students overwhelmed by competing commitments. Journal: *Nature*
- **Why environmental treaties must also address mental health** — score: 0.000 The article emphasizes the importance of integrating mental health considerations into environmental treaties, which have historically focused primarily on physical and ecological impacts. It argues that addressing the psychological and social consequences of climate change, biodiversity loss, and pollution is crucial for a comprehensive approach to environmental policy. Journal: *Nature*
- **This AI tool claims to pick the top 1% of preprints. Should researchers trust it?** — score: 0.000 The article examines an AI tool developed by QED Science that purports to identify the top 1% of preprints by evaluating them based on originality and validity, thereby aiming to minimize bias. It

raises questions about the reliability of this tool and whether researchers should place their trust in its assessments. Journal: *Nature*

- **Despite global turmoil, Gen Z researchers are not giving up** — score: 0.000 In the face of global challenges, four Generation Z researchers share their strategies for maintaining motivation and resilience in their work. They highlight personal stories and methods that help them navigate adversity while pursuing their research goals. Journal: *Nature*
- **Turning the tide: restoring Mozambican mangroves — in photos** — score: 0.000 The article highlights marine biologist Célia Macamo's efforts to preserve mangrove forests in Maputo, Mozambique, amid the pressures of urban development. It showcases her research aimed at restoring these vital ecosystems, which are crucial for coastal protection and biodiversity. Journal: *Nature*
- **In situ regeneration engineering enables low-cost and high-rate $\text{Na}_{2+2x}\text{Fe}_{2-x}(\text{SO}_4)_3$ cathodes from industrial byproducts** — score: 0.000 The article discusses a novel approach for engineering cathodes for sodium-ion batteries using in situ regeneration techniques, which allows for the creation of $\text{Na}_{2+2x}\text{Fe}_{2-x}(\text{SO}_4)_3$ from industrial byproducts. This method not only lowers production costs but also enhances the rate performance of the batteries, presenting a sustainable solution for energy storage. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Oxide electrolyte-driven interphase reconfiguration enables durable solid-liquid hybrid lithium batteries** — score: 0.000 The article discusses advancements in solid-liquid hybrid lithium batteries, focusing on the role of oxide electrolytes in facilitating interphase reconfiguration that enhances the durability of these energy storage systems. The authors highlight how this innovative approach can potentially improve battery performance and longevity. Journal: *ScienceDirect Publication: Energy Storage Materials*